

Syllabus for

Academic year 2021/2022 >

EEF031 - Theory of electromagnetic fields

Elektromagnetisk fältteori

Syllabus adopted 2021-02-26 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

7,5 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Main field of study: Electrical Engineering, Engineering Physics

Department: 32 - ELECTRICAL ENGINEERING

Teaching language: Swedish

Application code: 57139

Open for exchange students: No

Maximum participants: 180

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0194 Examination 7,5 c							Grading: TH 7,5 c Thu 13/01-2022 pm J, Tue 12/04-2022 pm J, Thu 18/08-2022 pm J

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 2 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 2 \(compulsory\)](#)

Examiner:

[Andreas Fhager](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Course specific prerequisites

Courses in mathematics.

Aim

To give ability to analyse and solve fundamental field problems.

Learning outcomes (after completion of the course the student should be able to)

To give ability to analyse and solve fundamental electromagnetic field problems.

Content

Electrostatics:

Charge and charge densities, Coulomb's law, electrostatic field, Gauss' law, electrostatic potential, conductors and insulators, electric dipoles and dipole fields, torque and forces on dipoles in electric fields, polarisation and polarisation charge densities, electric displacement, boundary conditions, capacitance calculations, electrostatic energy, energy density in the electric field, force calculation using the energy method, Poisson's and Laplace's equations, uniqueness theorem, electrostatic boundary value problems.

Steady electric current:

Current density, Ohm's law, equation of continuity, boundary conditions, relaxation time, Joule's law, resistance calculations.

Magnetostatics:

Magnetic flux density, Lorentz force, Ampère's law, magnetic vector potential, Biot-Savart's law, magnetic dipoles and dipole fields, torque and forces on dipoles in magnetic fields, magnetisation, magnetisation current densities, magnetic field intensity, boundary conditions, ferromagnetic hysteresis, inductance and mutual inductance, magnetic energy, energy density in magnetic field, force calculations using the energy method.

Electrodynamics:

Faraday's law of induction, displacement current density, Maxwell's equations, boundary conditions, wave equations, retarded potentials, complex vector fields, plane waves, skin effect, Poynting's theorem, reflection and transmission of plane wave at plane interface, Fresnel equations, Brewster angle, total internal reflection, antennas, Hertzian dipole.

Organisation

Lectures, tutorials, problem solving at home expected. A non-obligatory "mid-period exam" covering statics, in study week 4. Points from the mid-period exam can be used on respective part at the exam. If results are good at the mid-period exam the corresponding part at the exam can be skipped. Voluntary web-based hand-in questions every week can give bonus points on the exam.

Literature

Course book: DK Cheng: Field and Wave Electromagnetics (Pearson New International Edition)

Additional course material can be downloaded from the course webpage, <https://www.elmagn.chalmers.se/F2/>

Examination including compulsory elements

A written exam of five problems. Non-obligatory hand-in questions give bonus points. A voluntary "mid-period exam" on electrostatics and magnetostatics makes it possible to complete the course in two parts.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.